

Evolution of Network Package Routing

1. Cover Page:

- **The Evolution of Network Routing:**

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2. Introduction:

- **Brief Background:** The internet is an extremely complex system, and specifically we realized the routing of packages in particular makes beautiful use of various network protocols and graphs. Our project will delve into the logic behind 2 specific protocols and their place in the history of package routing.
 - **Motivation and Data Structures used :** The motivation behind this project is to create an interactive tool that demystifies the abstract concepts of package routing. By visually simulating different routing philosophies, we can understand not only *how* they work but *why* they were designed the way they were, and why modern protocols are so resilient. This entire field is fundamentally governed by Data Structures; the network itself is a **Graph**, routing decisions are stored and looked up in **Hash Tables**, and efficient path finding is achieved using **Priority Queues**. Our project aims to make these connections tangible.
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3. Problem Statement:

This project aims to design and develop an interactive, web-based simulator to visually demonstrate and compare two fundamental routing philosophies: **Distance-Vector** (exemplified by the Routing Information Protocol, RIP) and **Link-State** (exemplified by the Open Shortest Path First, OSPF protocol).

The scope of the project includes:

- Allowing a user to dynamically create a custom network topology (a graph of nodes and weighted edges).

- Implementing a turn-based simulation of the Distance-Vector protocol to show its slow, "routing-by-rumor" convergence.
 - Visually demonstrating the infamous "Counting to Infinity" problem, a critical failure mode inherent in the Distance-Vector approach.
 - Implementing a simulation of the Link-State protocol, showing how it builds a complete network map (LSDB) and uses Dijkstra's algorithm to rapidly and accurately calculate routes.
 - Providing a clear, side-by-side comparison that highlights the superiority and robustness of the modern Link-State approach.
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4. Objectives:

- To develop an intuitive visual tool that makes abstract routing algorithms accessible and understandable.
 - To implement and contrast the core logic of a Distance-Vector protocol (RIP) and a Link-State protocol (OSPF).
 - To create a live, demonstrable scenario showcasing the "Counting to Infinity" flaw.
 - To clearly illustrate the practical application and importance of Graph, Hash Table, and Priority Queue data structures in solving a real-world engineering problem.
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5. Literature Survey

The foundational principles for this project are drawn from established internet standards and academic literature. The Distance-Vector approach is formally defined in **RFC 1058**, which specifies the Routing Information Protocol (RIP). It describes the simple, hop-count-based metric and the process of routers advertising their entire routing tables to neighbors. In contrast, the Link-State approach is defined in **RFC 2328**, which details the Open Shortest Path First (OSPF) protocol. It specifies the use of Link-State Advertisements (LSAs), the "flooding" mechanism for distributing them, and the mandatory use of the Dijkstra algorithm for path calculation.

Standard networking textbooks, such as Kurose and Ross's "Computer Networking: A Top-Down Approach," compare these two philosophies directly. They highlight the trade-offs: RIP's simplicity and low computational overhead versus OSPF's complexity, speed, and robustness. The "Counting to Infinity" problem is a well-documented limitation of RIP discussed in these texts. While many resources describe these protocols, a research gap exists for interactive tools that allow a user to visually compare both protocols' dynamic behavior and failure modes on the same user-defined network topology. This project aims to fill that gap.

6. Relevance / Societal Impact:

The primary societal impact of this project is **educational**. It transforms a complex, invisible process into an interactive and engaging learning experience. By allowing students to "break" the network and see the consequences, it provides a deeper intuition for network engineering principles. The **novelty** of our approach lies not in implementing a single algorithm, but in crafting a narrative of technological evolution. By contrasting the flawed "rumor-based" system with the robust "map-based" system, we tell a compelling story about why modern networks are designed the way they are.

7. Approach & Methodology

Our approach follows a narrative to guide the user through the evolution of routing.

Flowchart of User Interaction:

- **Act 1: The Past (RIP Works):** A user builds a network. They run the RIP simulation and watch as the routers slowly converge on the correct paths through a round-based "gossip" mechanism.(We also explain the necessity to use RIP back in the day (memory limitations))
- **Act 2: The Failure (RIP Breaks):** The user simulates a link failure. Running the RIP simulation again reveals the "Counting to Infinity" problem, where routers create a routing loop, unable to correct themselves.
- **Act 3: The Present (OSPF Succeeds):** On the same broken network, the user switches to the OSPF simulation. They witness the rapid "flooding" of LSAs, the creation of a complete network map, and the immediate calculation of a new, correct path via Dijkstra's algorithm.

Data Structures Employed:

- **Graph (Adjacency List):** The natural choice to represent the network. The list structure is efficient for finding the neighbors of any given router.
 - **Hash Table:** Used to implement the routing table in RIP and the Link-State Database in OSPF. This provides efficient $O(1)$ average time complexity for lookups, insertions, and updates, which is critical for the simulation's performance.
 - **Priority Queue (Min-Heap):** Used to optimize the implementation of Dijkstra's algorithm within the OSPF simulation, reducing its time complexity from $O(V^2)$ to $O(E \log V)$.
 - Some other minor data structures such arrays will also be used
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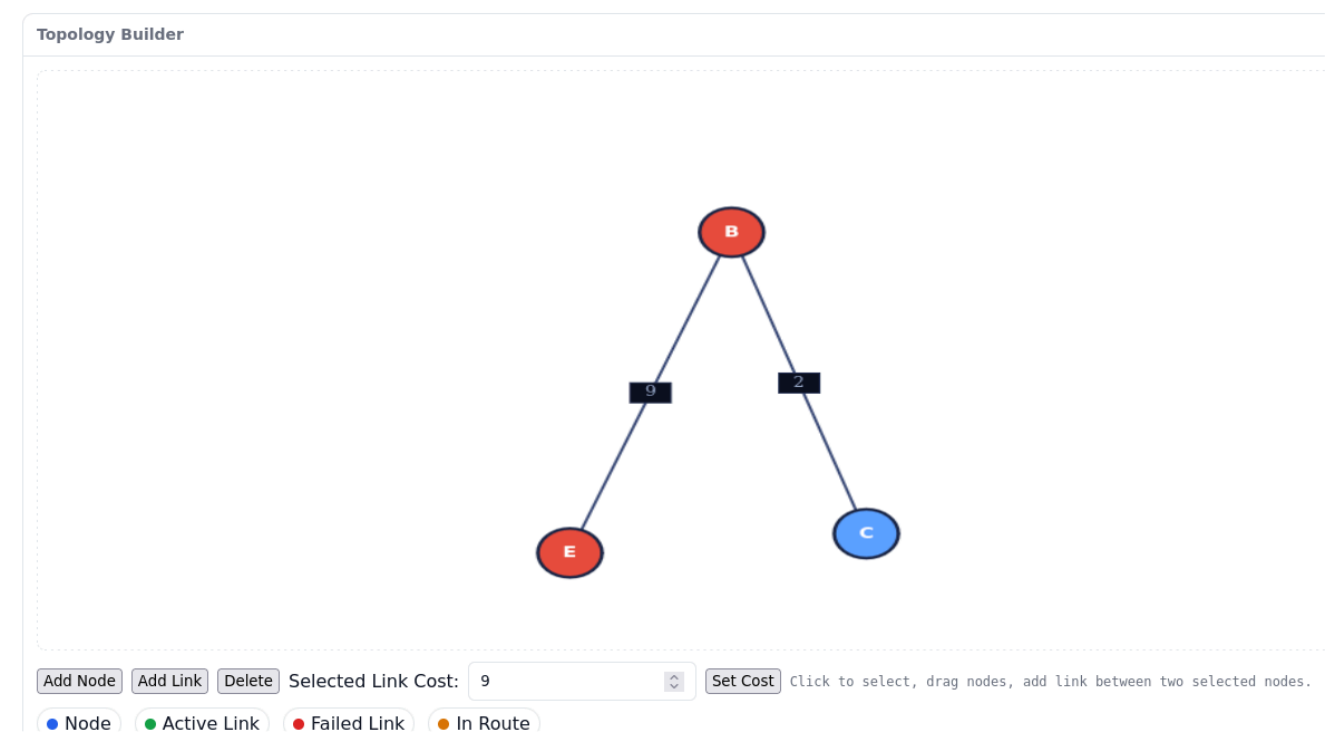
8. Current Status of Work

- **Completed Tasks:**
 - We've studied the core principles involved in package routing and also, detailed study on the protocols we're about to implement

- Initial Designs of how the interface will look are complete, added some basic React components, such as buttons , a way to add nodes etc, and chose appropriate js libraries.
- The logic for the Distance-Vector (RIP) protocol, including the round-based update mechanism, is implemented in cpp currently we're bring it over to js to connect it to our front end
- **Pending Tasks:**
 - Finalize the visualization for the "Counting to Infinity" scenario.
 - True implementation of the algorithms and connecting them with front end
 - Implement the full Link-State (OSPF) protocol, including LSA flooding and the Dijkstra's algorithm module.
 - Integrate the two simulations into a single, seamless user experience.
 - Prepare the final report, ppt and demo video
- **Challenges Faced:**
 - Primary challenge was finding a way to visualize the data structures used clearly, along with making the front end actually use the algorithm (we know how the algorithm works, how to write just the algorithm but need to visualization to be actually based on this algorithm)

9. Implementation Details

- **Programming Language:** JavaScript (ES6), React, tailwind css
- **Environment:** VS Code, nvim (anupam)



Our intial designs, still need to style things up

10. Conclusion & Future Scope

The project is on track, with the foundational data structures and the complete Distance-Vector simulation forming a solid base. The current implementation already allows for the creation and visualization of dynamic networks. The next and final phase will focus on implementing the more complex Link-State protocol to complete the project's comparative narrative, properly integrating the algorithms with the visualizations.

Upon completion, a possible future extension would be to implement a simplified version of **BGP (Border Gateway Protocol)**. This would add a "fourth act" to our evolution story, showing how routing is handled *between* large networks based on policy rather than just speed, providing an even more complete picture of how the internet works.

It's not all just a simple routing problem, there might be paid routes (negative weights) and there's also laws about routes to be used, BGP accounts for real world conditions, however was too complicated and had too many variables to account for.

11. References

- Malkin, G. (1988). *RFC 1058: Routing Information Protocol*. IETF.
- Moy, J. (1998). *RFC 2328: OSPF Version 2*. IETF.
- Kurose, J. F., & Ross, K. W. (2021). *Computer Networking: A Top-Down Approach* (8th ed.). Pearson.